POLYLOGARITHM VALUES AT A GOLDEN RATIO-BASED ARGUMENT

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ABSTRACT. This paper introduces and analyzes a specific complex constant, denoted Λ_{G1}, which is derived from fundamental properties of the golden ratio, ϕ. We define the constant's components based on inverse powers of ϕ and prove the key property that its magnitude is less than one. This validates the use of Λ_{G1} as an argument in the Polylogarithm function, $\mathrm{Li}_s(z)$. We present high-precision numerical evaluations for the Dilogarithm ($s = 2$) and Trilogarithm ($s = 3$) of this constant. Based on this numerical evidence, we conjecture that the values $\mathrm{Li}_s(\Lambda_{G1})$ are transcendental for all integers $s \geq 2$ and do not belong to the field extension $\mathbb{Q}(\pi, e, \ln(2), \phi, \gamma)$.

1. Introduction

The study of special functions and constants often reveals connections between different areas of mathematics. This paper focuses on the properties of a specific complex constant derived from the golden ratio, $\phi = (1+\sqrt{5})/2$. The motivation for this constant stems from the classical problem of dividing a line segment in extreme and mean ratio, as found in Euclid's *Elements* [2]. We define the components of our constant directly from inverse powers of ϕ, providing a non-arbitrary foundation for its construction. The purpose of this paper is to rigorously define this constant, Λ_{G1}, and establish its validity as an argument in the Polylogarithm function. This allows us to define and analyze the values of $\mathrm{Li}_s(\Lambda_{G1})$, which are the central topic of this work. Understanding the nature of these specific values may offer insights into mathematical domains where the golden ratio is foundational. For instance, given that the diffraction patterns of quasi-crystals are intrinsically linked to the golden ratio, exploring the analytic properties of values like $\mathrm{Li}_s(\Lambda_{G1})$ may prove to be a fruitful direction. The paper culminates in a formal conjecture about the transcendental nature of these values, based on high-precision numerical analysis.

2. The Constant Λ_{G1} and its Foundational Properties

2.1. Defining Components from the Golden Ratio.

We define two real constants, T and J, directly from the algebraic properties of the golden ratio.

Definition 2.1 (Primary Real Components)**.** The constants T and J are defined from the golden ratio, ϕ, as:

$$T = \frac{1}{2\phi} \quad \text{and} \quad J = \frac{1}{2\phi^2}$$

These definitions lead to a pair of linear relations that highlight their connection to classical geometry.

Proposition 2.2 (System of Equations)**.** *The constants T and J are the unique real numbers (a, b) that satisfy the system:*

$$a/b = \phi \tag{1}$$

$$a + b = 1/2 \tag{2}$$

Proof. The ratio is immediate from the definitions: $T/J = (1/(2\phi))/(1/(2\phi^2)) = \phi^2/\phi = \phi$. For the sum, we use the identity $\phi + 1 = \phi^2$:

$$T + J = \frac{1}{2\phi} + \frac{1}{2\phi^2} = \frac{\phi+1}{2\phi^2} = \frac{\phi^2}{2\phi^2} = \frac{1}{2}$$

The system has a unique solution because the determinant of the corresponding linear system is non-zero. □

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Remark. This formulation shows that the constraint $T + J = 1/2$ is not an arbitrary choice, but a direct consequence of defining the components from successive inverse powers of ϕ. As discussed in Appendix A, this corresponds to the classical geometric problem of dividing a line segment of length $L = 1/2$ in the golden ratio. The explicit values are $T = (\sqrt{5} - 1)/4$ and $J = (3 - \sqrt{5})/4$.

From these constants, we define our primary object of study.

Definition 2.3 (The Primary Constant)**.** The constant Λ_{G1} is defined as the complex number $\Lambda_{G1} = T + iJ$.

2.2. **Convergence Property.** A crucial property of Λ_{G1} is that its magnitude is less than one, which is required for the convergence of the Polylogarithm series.

Lemma 2.4 (Magnitude of Λ_{G1})**.** *The constant Λ_{G1} lies within the unit disk, i.e., $|\Lambda_{G1}| < 1$.*

Proof. We require $|\Lambda_{G1}|^2 < 1$. We calculate the squared values of the components: $T^2 = \left(\frac{\sqrt{5}-1}{4}\right)^2 = \frac{6-2\sqrt{5}}{16}$ and $J^2 = \left(\frac{3-\sqrt{5}}{4}\right)^2 = \frac{14-6\sqrt{5}}{16}$. Summing these gives:

$$\begin{aligned} |\Lambda_{G1}|^2 = T^2 + J^2 &= \frac{6 - 2\sqrt{5} + 14 - 6\sqrt{5}}{16} \\ &= \frac{20 - 8\sqrt{5}}{16} = \frac{5 - 2\sqrt{5}}{4} \approx 0.131966 \end{aligned}$$

Since $0.131966 < 1$, the condition is satisfied. □

3. Defining the Polylogarithm Values of Λ_{G1}

The property $|\Lambda_{G1}| < 1$ is central to this paper, as it allows us to define the Polylogarithm values at this constant. The Polylogarithm function is a key object in this study.

Definition 3.1 (The Polylogarithm Function)**.** For any complex s, the Polylogarithm function of order s, $\mathrm{Li}_s(z)$, is defined by the power series:

$$\mathrm{Li}_s(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^s}$$

This series converges for all complex numbers z such that $|z| < 1$.

Since Λ_{G1} satisfies the convergence condition, we can define the specific numerical values that are the subject of our main conjecture:

$$\mathrm{Li}_s(\Lambda_{G1}) = \sum_{n=1}^{\infty} \frac{(\Lambda_{G1})^n}{n^s}$$

This identification holds for any complex s. Furthermore, the convergence property allows for the exact evaluation of other related series, such as the polynomial-weighted power series discussed in Appendix B.

4. Numerical Analysis and a Conjecture on the Nature of $\mathrm{Li}_s(\Lambda_{G1})$

4.1. **Numerical Values.** We present high-precision numerical evaluations for the first two non-trivial integer cases of the Polylogarithm of Λ_{G1}. The values were computed using the Python ‘mpmath‘ library with a working precision of 200 decimal digits to ensure robustness.

- For $s = 2$, the Dilogarithm of Λ_{G1} is:

$$\mathrm{Li}_2(\Lambda_{G1}) \approx 0.322328253720993 + 0.226730769307749i$$

- For $s = 3$, the Trilogarithm of Λ_{G1} is:

$$\mathrm{Li}_3(\Lambda_{G1}) \approx 0.310345112613931 + 0.198884393433621i$$

4.2. **A Conjecture on the Nature of These Values.** The transcendental nature of special function values at algebraic arguments is a central topic in number theory [1]. For context, the values of the Dilogarithm are known to be transcendental at certain related real algebraic points, such as $\mathrm{Li}_2(1/2) = \frac{\pi^2}{12} - \frac{1}{2}\ln^2(2)$ and $\mathrm{Li}_2(\phi^{-2}) = \frac{\pi^2}{15} - \ln^2(\phi)$ [3]. The constant Λ_{G1}, being a non-real algebraic number that is not a root of unity, presents a distinct and challenging case that falls outside the scope of many established techniques. Our high-precision values could not be resolved into a closed-form expression by the PSLQ integer relation algorithm using a standard basis of constants (e.g., powers of $\pi, \ln(2), \phi$). While not a formal proof, this failure provides numerical motivation for their apparent transcendental nature and leads us to propose the following general conjecture.

Conjecture 4.1. *For any integer $s \geq 2$, the value $\mathrm{Li}_s(\Lambda_{G1})$ is a transcendental number. Furthermore, it is conjectured that this number is not in the field $\mathbb{Q}(\pi, e, \ln(2), \phi, \gamma)$.*

Remark. Proving this conjecture is beyond the scope of this paper, but proposing it frames a clear direction for future work.

5. Conclusion

This paper has introduced a complex constant, Λ_{G1}, whose definition is derived non-arbitrarily from the golden ratio. We have proven that its magnitude is less than one, which validates its use as an argument in the Polylogarithm function. The primary contribution of this work is the introduction of this constant and the subsequent analysis of its polylogarithmic values, $\mathrm{Li}_s(\Lambda_{G1})$. Based on high-precision numerical evidence from the $s = 2$ and $s = 3$ cases, we have proposed a new open problem: a conjecture on the transcendental nature of these values.

6. Future Work

The conjecture presented in Section 4 opens a clear path for future investigation. A primary goal remains to prove or disprove the transcendental nature of $\mathrm{Li}_s(\Lambda_{G1})$. More specific avenues for future work include:

- Analyzing the real and imaginary parts of $\mathrm{Li}_s(\Lambda_{G1})$ separately to determine if they possess any distinct algebraic properties.
- Exploring other properties of the constant Λ_{G1} itself.
- Investigating potential connections to other fields where the golden ratio is fundamental. As noted, the role of ϕ in quasi-crystals suggests this as a promising area of inquiry.

Appendix A. Geometric Motivation for the Defining Equations

The definitions of T and J have a direct geometric motivation arising from the division of a line segment according to the golden ratio. If a segment of length L is divided into two pieces, a and b, such that $a/b = \phi$, the pieces have lengths $a = L/\phi$ and $b = L/\phi^2$. Our constants T and J correspond to this construction for $L = 1/2$. As shown in Proposition 2.2, this choice of $L = 1/2$ is the natural sum that arises when the components T and J are defined from the fundamental quantities $1/(2\phi)$ and $1/(2\phi^2)$. This provides a non-arbitrary basis for the constant's definition.

Appendix B. Evaluation of a Related Polynomial-Weighted Series

As a consequence of Lemma 2.4, Λ_{G1} is a valid argument in the series whose sum is given by Eulerian polynomials, $A_k(x)$. This provides a direct evaluation for a family of series with Λ_{G1} as the base. For any integer $k \geq 0$:

$$\sum_{n=0}^{\infty} n^k \cdot (\Lambda_{G1})^n = \frac{A_k(\Lambda_{G1})}{(1 - \Lambda_{G1})^{k+1}}$$

While not central to the paper's main conjecture, this identity is an interesting property of the constant Λ_{G1}.

References

[1] A. Baker. *Transcendental Number Theory*. Cambridge University Press, 1975.

[2] Euclid, et al. *The Thirteen Books of Euclid's Elements*. In *Great Books of the Western World*, vol. 11, edited by R. M. Hutchins, Encyclopaedia Britannica, 1952.

[3] L. Lewin. *Polylogarithms and Associated Functions*. North-Holland, 1981.